

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA0202

COURSE OUTCOME COVERED

CO2: Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 2

Total Marks:20

Annexure A

Class Test

Code of Conduct:

I P Sai Bhavyatha(192424333) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Palacharla Sai Bhavyatha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	

Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	
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Annexure A

Class Test

Question 1: Image Preprocessing and Feature Extraction

Scenario: A drone is used to capture aerial images of a farmland to identify pestaffected crop regions. Due to sensor limitations, wind movement, and changing sunlight, the captured images contain noise, shadows, and uneven illumination. Pestaffected areas usually show visible texture and color variations compared to healthy crop regions.

Answer the following questions:

a) Preprocessing Pipeline — 4 Marks

Propose a suitable preprocessing pipeline to improve the quality of the farmland images before analysis. Your answer should include methods for noise reduction, illumination correction, contrast enhancement, and normalization. Justify the purpose of each step.

b) Feature Extraction — 3 Marks

Choose an appropriate feature extraction method to identify texture changes caused by pest damage. Explain why the selected method is suitable for detecting affected crop regions.

c) Rotation and Scale Variations — 3 Marks

Drone images may be captured from different heights and angles. Discuss how rotation and scale variations can affect feature detection. Suggest suitable methods to make the feature extraction process more robust.

Question 2: Stereo Vision and Motion Estimation

Scenario: A self-driving car is equipped with stereo cameras to navigate through a busy street. The system must estimate the distance of nearby vehicles and pedestrians in real time. It should also track moving objects accurately to avoid collisions, especially when pedestrians or vehicles change speed suddenly.

Answer the following questions:

a) Stereo Disparity and Depth Estimation — 4 Marks

Explain how stereo cameras use the difference between left and right image views to estimate the depth of objects. Discuss how disparity changes for nearby and distant objects.

b) Optical Flow for Motion Tracking — 3 Marks

If a pedestrian suddenly changes speed or direction, explain how optical flow helps in tracking the pedestrian's motion between consecutive video frames. c) Errors and Mitigation Techniques — 3 Marks

Identify possible sources of error in depth estimation and motion tracking, such as camera misalignment, poor lighting, occlusion, fast motion, or textureless regions.

Suggest suitable techniques to reduce these errors.

Question 1: Image Preprocessing and Feature Extraction

a) Preprocessing Pipeline (4 Marks)

A suitable preprocessing pipeline for farmland images is:

1. Noise Reduction

- Apply **Median Filter** or **Gaussian Filter**.
- Removes sensor noise and reduces unwanted disturbances while preserving important image details.

2. Illumination Correction

- Use **Histogram Equalization** or **Adaptive Histogram Equalization (CLAHE)**.
- Corrects uneven lighting caused by changing sunlight and shadows.

3. Contrast Enhancement

- Apply **CLAHE** or contrast stretching.
- Improves visibility of pest-affected crop regions by making texture and color differences more noticeable.

4. Normalization

- Normalize pixel values to a fixed range (0–1 or 0–255).
- Ensures consistent image intensity for accurate feature extraction and comparison.

Preprocessing Pipeline

Input Image

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Noise Reduction (Median/Gaussian Filter)

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Illumination Correction (Histogram Equalization/CLAHE)

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Contrast Enhancement

↓

Normalization

↓

Preprocessed Image

b) Feature Extraction (3 Marks)

A suitable feature extraction method is the **Local Binary Pattern (LBP)**.

Reason:

- LBP effectively captures **texture patterns**.
- Pest-affected crops show rough, damaged, or irregular textures compared to healthy crops.
- It is computationally efficient and works well under varying lighting conditions.

Therefore, LBP is suitable for detecting pest-damaged crop regions.

c) Rotation and Scale Variations (3 Marks)

Drone images are captured from different heights and viewing angles.

Effects

- Rotation changes the orientation of image features.
- Scale changes the size of crop regions.
- These variations may cause incorrect or missed feature detection.

Robust Methods

- **SIFT (Scale-Invariant Feature Transform)**: Handles both scale and rotation changes.
 - **SURF (Speeded-Up Robust Features)**: Faster alternative with similar robustness.
 - Image augmentation or multi-scale analysis can further improve detection accuracy.
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Question 2: Stereo Vision and Motion Estimation

a) Stereo Disparity and Depth Estimation (4 Marks)

Stereo cameras consist of two cameras placed at a fixed distance apart.

Both cameras capture the same scene from slightly different viewpoints.

The horizontal difference in the position of the same object in the two images is called **disparity**.

Depth Estimation

Depth is calculated using:

$$\text{Depth} = (\text{Focal Length} \times \text{Baseline Distance}) / \text{Disparity}$$

Relationship Between Disparity and Distance

- **Nearby objects** → Large disparity → Smaller depth value (closer).
- **Distant objects** → Small disparity → Larger depth value (farther).

This allows the self-driving car to estimate the distance of vehicles and pedestrians accurately.

b) Optical Flow for Motion Tracking (3 Marks)

Optical Flow estimates the movement of pixels between consecutive video frames.

When a pedestrian suddenly changes speed or direction:

- The motion vectors change accordingly.
- Optical flow continuously tracks the new position.
- It predicts movement in real time.
- The vehicle can update its path and avoid collisions.

Thus, optical flow enables reliable motion tracking of moving objects.

c) Errors and Mitigation Techniques (3 Marks)

Possible Errors

- Camera misalignment
- Poor lighting conditions

- Occlusion (objects blocking each other)
- Fast object movement
- Textureless surfaces

Mitigation Techniques

- Perform accurate **camera calibration** to correct alignment.
- Use **adaptive illumination enhancement** for low-light conditions.
- Apply **feature-based matching algorithms** such as SIFT or ORB.
- Use **Kalman Filters** or **Particle Filters** for smoother motion tracking.
- Employ **deep learning-based stereo matching** to improve depth estimation in difficult scenes.

These methods improve both depth estimation and motion tracking accuracy.

Conclusion

Image preprocessing improves image quality by reducing noise, correcting illumination, enhancing contrast, and normalizing pixel values. Feature extraction methods such as LBP effectively identify pest-damaged crop regions. Stereo vision estimates object depth using disparity, while optical flow tracks moving objects between frames. Techniques such as camera calibration, robust feature extraction, and motion filtering help reduce errors and improve the performance of intelligent vision systems.